

Article

Economic Evaluation Method of Modern Power Transmission System Based on Improved Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) and Best-Worst Method-Anti-Entropy Weight

Wenhui Zeng ¹, Jiayuan Fan ², Zhichao Ren ¹, Xiaoyu Liu ¹, Shuang Lv ³, Yuqian Cao ^{4,*}, Xiao Xu ⁴ and Junyong Liu ⁴

¹ State Grid Sichuan Economic Research Institute, Chengdu 610095, China

² State Grid Sichuan Information & Communication Company, Chengdu 610299, China

³ State Grid Sichuan Electric Power Company Chengdu Power Supply, Chengdu 610041, China

⁴ School of Electrical Engineering, Sichuan University, Chengdu 610017, China

* Correspondence: caopinno@gmail.com

Abstract: As the demand for power supply increases, the investment in the power transmission system constantly increases. An accurate economic evaluation of the power transmission system is essential for future investment decisions and management. Applying a single method in economic evaluation leads to excessive subjective consciousness and unreasonable weight allocation. The Euclidean distance in the traditional TOPSIS method only partially works on the condition that the criteria are linearly correlated. To solve these problems, an economic evaluation method based on improved TOPSIS and BWM-anti-entropy weight is proposed. For the assignment of weights, the method retains the advantages of subjective and objective weighting methods based on the Nash equilibrium, breaks through the limitation of utilizing a single method, which contributes to one-sided results, and enhances the scientific rigor and rationality of the comprehensive weighting process. Furthermore, based on comprehensive weights, the method improves the TOPSIS by introducing the Mahalanobis distance and Pearson correlation coefficients, which can eliminate the influence of linear correlation. Finally, ten 500 kV transmission and transformation projects are analyzed and ranked to verify the method's feasibility. Empirical analysis shows that the method can effectively evaluate the economic benefits of the power transmission system.

Keywords: power transmission system; economic evaluation; best-worst method (BWM); anti-entropy weight method; Nash equilibrium; improved TOPSIS



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1. Introduction

Energy and environmental problems are becoming increasingly prominent, and electrical energy is an indispensable secondary energy source in society. As the industry has progressed in recent years, electricity consumption has steadily increased, resulting in elevated demands for power supply reliability and safety. The power transmission system is an integral part of the electric power system—which includes transmission lines, substations, and related equipment—influencing the quality of power supply and economic and social development. The power transmission system has a significant investment scale and a long construction period [1]. Therefore, the economics of power system projects in terms of profitability, liabilities, etc. need to be quantified and comprehensively evaluated for managerial decision-making and planning.

A comprehensive evaluation uses scientific, feasible methods to evaluate and analyze all aspects of the evaluation: object behavior, multi-indicator, and multi-level. Some studies have concluded that the economic evaluation of the power transmission system focuses

on cost and benefit. Based on the cost–benefit analysis, the scores for multiple power transmission systems can be calculated and ranked [2]. According to the characteristics of the power system project, the economic evaluation of the project is divided into two parts: financial analysis and economic cost-effectiveness analysis [3]. Subsequently, increasing numbers of people are researching and improving the evaluation system and criteria for power transmission systems. A two-tier indicator system for investment effectiveness evaluation was established, considering both unit- and macro-investment effectiveness [4]. To combine the metrics of investment effectiveness and investment efficiency, the criteria of comprehensive economic evaluation can be categorized into basic, modified, and appraisal criteria [5]. Existing research provides an important reference for the economic evaluation of the power system in this paper. This paper considers the three perspectives of time, value, and efficiency to establish a comprehensive evaluation system from the three aspects of profitability, solvency, and development capability.

Regarding models and methods for the economic evaluation of power transmission systems, many relevant research studies already exist in the field of MCDA. Multi-criteria decision analysis (MCDA) is a complex field of decision-making disciplines for problems with multiple criteria and multiple choices that can be used for economic evaluation [6]. The core methods and techniques are WSM, TOPSIS, AHP, PROMETHEE, ELECTRE, COMET, and SIMUS [7]. The weighted sum method (WSM) is the most commonly used [8] for assessment and decision-making in a single dimension. The analytical hierarchy process (AHP) uses a hierarchical structure to decompose a complex problem to evaluate the objectives at the top of the structure [9]. The ELECTRE method can be iterated to provide a ranking of alternatives based on consistency and thresholds [10]. The combined approaches are also proposed, and some previous research conducted a comprehensive analysis by establishing a fuzzy synthetic evaluation model using the hierarchical analysis method [11]. Pareto optimal function and merit criterion methods are also employed to enhance project comparison and decision-making [12,13]. In the study [14], which optimizes the fuzzy synthetic evaluation model, a hierarchy evaluation model for fuzzy analysis is built based on a novel comprehensive fuzzy evaluation operator. However, this model relies too much on subjectivity. A combination of the entropy weighting method and gray correlation analysis is employed to address the challenges posed by incomplete information and the subjective nature of weights. Then, the improved gray correlation analysis method is used to arrive at the optimal solution [15]. Based on the content and characteristics of specific projects in the power system, this paper proposes the improved TOPSIS method. TOPSIS is a multi-criteria decision analysis method that can be used to evaluate objects across multiple areas with satisfactory results [16]. The original TOPSIS method is likely to lead to the problem of rank reversal when changing the number of items to be evaluated [17], and the TOPSIS method can be improved by introducing the absolute ideal solution [18,19]. The absolute ideal solution provides initial support for our improvements to TOPSIS. We find that indicators are often correlated, and that correlation leads to partial failure of the Euclidean distance in traditional TOPSIS [20,21]. Although many methods and software have been applied to build evaluation systems and accurately evaluate the economics of power transmission systems, few weights can balance subjective experience and objective data.

Previous studies have often used single-dimension weights to evaluate economics, with some articles considering only subjective weights [22,23] (AHP, BWM), and some articles considering only objective weights [24,25] (EWM, PCA). Combinations of subjective and objective methods are also often used in the field of evaluation [26,27]. The construction of the coupling coordination model is dependent on the coordination of weights estimated by AHP and EWM methods [28]. In the paper [29], a novel hybrid multi-criteria decision-making (MCDM) approach is devised for offshore wind turbine selection by assignment of weights using ANP and EWM. Given the limitations of assigning weights to a single dimension, we propose the BWM-anti-entropy weight method in this paper, which utilizes

a game theory model to balance subjective and objective weights. Game theory models can ultimately result in more efficiently comprehensive weights [30].

In summary, based on comprehensive weights and absolute ideal solutions, the TOPSIS method can be improved using the Mahalanobis distance and Pearson's correlation coefficient. This final improved TOPSIS method is more scientific and logical.

The key processes of the improved TOPSIS method are illustrated in Figure 1.

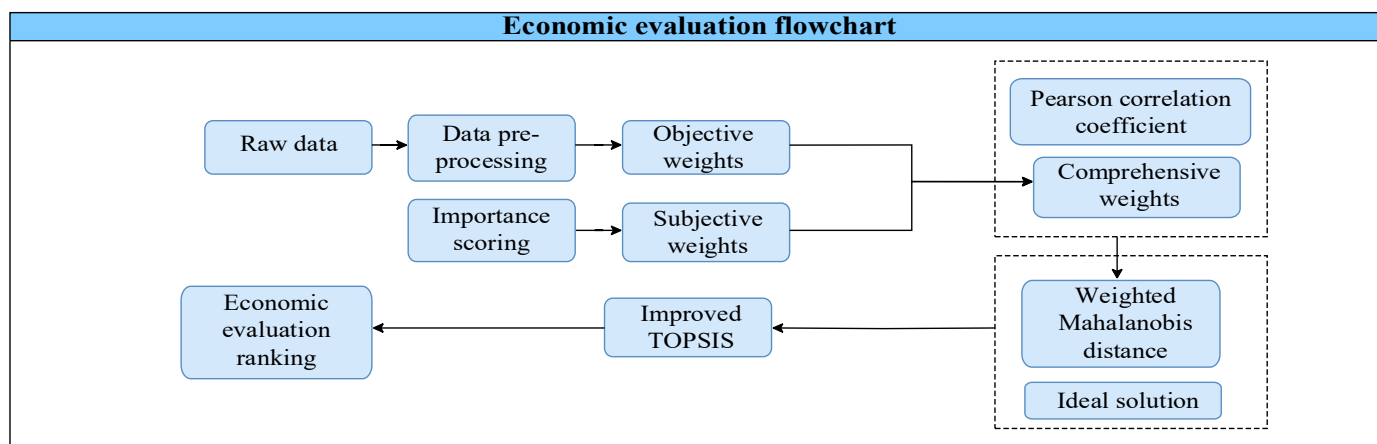


Figure 1. Economic evaluation flowchart.

Other contents are organized as follows: The economic evaluation system of power transmission systems is constructed in Section 2, focusing on three aspects and determining the specific evaluation criteria. In Section 3, the subjective weights and objective weights are calculated, then an integrated weighting model is established by applying the game theory. Section 4 outlines the deficiencies of the traditional TOPSIS method and introduces the improved TOPSIS method utilizing the Mahalanobis distance and Pearson coefficient. In Section 5, the comparison analysis is carried out on the specific data of 500 kV transmission and transformation projects to acquire the evaluation results for the proposed criterion system, model, and improved method.

2. The System of Criteria for Economic Evaluation

The comprehensive evaluation of economic efficiency in this paper is based on the financial evaluation criterion system.

According to the characteristics of the power transmission system and the theory of financial evaluation criteria, this paper establishes the evaluation system from three perspectives: financial profitability, financial solvency, and financial development capability [31]. The evaluation of comprehensive economic efficiency is carried out mainly from the perspectives of time, value, and efficiency [32,33], and these three perspectives are substantially equivalent, which means that many of the criteria are consistent at their core.

Under each perspective, the appropriate criteria should be selected for the description. The information on different criteria is used for the economic evaluation of the power transmission system. The specific criteria system is shown in Table 1.

Table 1. Economic evaluation criteria system.

Level Target	Level 1 Criteria	Level 2 Criteria	Criteria Code	Criteria Type
Economic evaluation method of the power transmission system	Financial Profitability	Return on Investment	C ₁	Forward
		Net Present Value	C ₂	Forward
		Payback Period	C ₃	Backward
		Internal Rate of Return	C ₄	Forward

Table 1. Cont.

Level Target	Level 1 Criteria	Level 2 Criteria	Criteria Code	Criteria Type
	Financial Solvency	Asset Liability Ratio	C ₅	Backward
		Current Ratio	C ₆	Intermediate Value
	Financial Development Capability	Profit Growth Rate	C ₇	Forward

1. Return on Investment (ROI)

ROI is a performance assessment used to evaluate the efficiency of an investment or to compare the efficiency of many different investments. ROI directly measures the amount of return on a particular investment relative to the cost of the investment. The ratio and profitability of the power transmission system have a significant positive correlation. The calculation formula is defined as shown in Equation (1) [34]:

$$ROI = \frac{A}{I} \times 100\% \quad (1)$$

where A is Annual net income, I is total investment (including construction investment and working capital).

2. Net Present Value (NPV)

Net present value (NPV) is the equivalent present value of the net cash flow for each year of the power transmission system life cycle discounted at a given discount rate, calculated as shown in Equation (2) [34]:

$$NPV = \sum_{t=0}^n (CI - CO)(1 + i_c)^{-t} \quad (2)$$

where CI is cash inflows by year, CO is cash outflows by year, n is the whole life cycle of the power transmission system, including the construction period and operation period.

3. Payback Period (P_t)

The payback period is the time required to recover the entire investment in a power transmission system in terms of its net income, usually in years. The payback period ends when a transmission system's net benefits equal the total initial investment. This number of years to recover the investment. This paper uses a static payback period without considering the time value. The formula is expressed in Equation (3) [34]:

$$\sum (CI - CO)_t = 0 \quad t = 1 \dots P_t \quad (3)$$

where $CI - CO$ is net cash flow.

4. Internal Rate of Return (IRR)

The internal rate of return (IRR) is the discount rate when the cumulative present value of the net cash flow over the life cycle of the power transmission system is zero. Internal rate of return (IRR) is the main dynamic evaluation criterion used to examine the profitability of the power transmission system, reflecting the profitability of the whole investment during its life. This is given by Equation (4) [34]:

$$\sum_{t=0}^n (CI - CO)(1 + IRR)^{-t} = 0 \quad (4)$$

where $CI - CO$ is net cash flow, n is the whole life cycle of the power transmission system, including the construction period and operation period.

5. Asset Liability Ratio (ALR)

Asset liability ratio refers to the percentage of total liabilities to total assets, reflecting the degree of financial risk and the liabilities of power transmission systems. This is given by Equation (4) [34]:

$$ALR = \frac{TL}{TA} \times 100\% \quad (5)$$

where TL is total liabilities, TA is total assets.

6. Current Ratio (CR)

The current ratio reflects the ability of the power transmission system to repay the current outstanding liabilities each year. It is an essential indicator of the short-term solvency of the power transmission system. The calculation formula [34] is:

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}} \times 100\% \quad (6)$$

7. Profit Growth Rate

The profit growth rate refers to the growth rate of the net profit of the power transmission system in the current period compared with the net profit in the previous period, and the larger the indicator's value, the stronger the profitability [34].

$$\text{Profit Growth Rate} = \frac{\text{Net profit for the period}}{\text{Net profit of the previous period}} \times 100\% \quad (7)$$

The economic evaluation criteria system is shown in Figure 2.

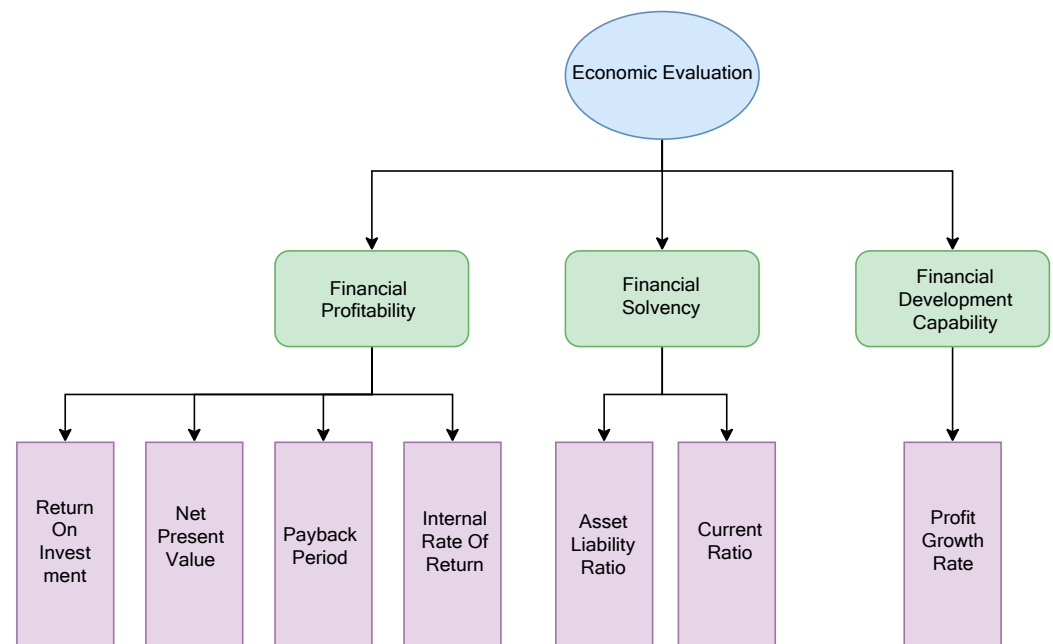


Figure 2. Economic evaluation criteria system.

3. Comprehensive Weighting Solution Model

3.1. Subjective Weight Solving Based on the BWM Method

The BWM is proposed to determine the subjective weights of the criteria [35]. In the past, the most common method was the AHP method, which compares any two indicators with each other to determine the evaluation matrix of the indicators. This requires $n(n-1)/2$ comparisons to determine the $n^2 - n$ data [36]. Compared to the AHP method, the BWM is less cumbersome and data-intensive, reducing errors caused

by confused thinking. The BWM makes it easier to pass the consistency test, making the results more reliable [37].

The BWM (best-worst method) is frequently used in multi-criteria decision-making [38]. Experts select the most important and least important criteria based on experience and practical engineering needs, then compare the best criterion with the rest of the criteria, and then compare the worst criterion with the rest. An integer value of 1–9 reflects the relative significance among the indicators. The specific procedures are as follows:

STEP 1: Identify the influencing factors of multi-criteria evaluation issues and establish a set of criteria $\{c_1, c_2, \dots, c_n\}$ based on the influencing factors of different aspects.

STEP 2: Identify the most important criterion and the least important criterion.

STEP 3: Compare the best, worst, and other criteria separately to create two vectors that characterize relative preferences. $A_B = (a_{B1}, a_{B2}, \dots, a_{Bn})$, $A_W = (a_{1W}, a_{2W}, \dots, a_{nW})$.

For the criteria system of the power transmission system, two scoring tables are established, as shown in Tables 2 and 3.

Table 2. Scoring the relative importance of the best criterion.

	c_1	c_2	c_3	$\dots$	c_7
C_B	a_{B1}	a_{B2}	a_{B3}	$\dots$	a_{B7}

Table 3. Scoring the relative importance of the worst criterion.

	c_1	c_2	c_3	$\dots$	c_7
C_W	a_{1W}	a_{2W}	a_{3W}	$\dots$	a_{4W}

STEP 4: Build a mathematical subjecting model and find the best solution to obtain the objective optimal weights. The mathematical planning [35] is as follows:

$$\text{s.t.} \begin{cases} \left| \frac{\omega_B}{\omega_j} - a_{Bj} \right| \leq l \\ \left| \frac{\omega_j}{\omega_W} - a_{jW} \right| \leq l, j = 1, 2, \dots, n \\ \sum_{j=1}^n \omega_j = 1 \\ \omega_j \geq 0 \end{cases} \quad (8)$$

where ω_B is the weight of C_B , ω_j is the weight of the base vector C_j , i.e., the actual weight of the criteria, ω_W is the weight of C_W , a_{Bj} is the value of the importance of C_B relative to C_j , a_{jW} is the value of the importance of C_j relative to C_W .

STEP 5: Use the following formula [35] to calculate C_R ; $C_R = 0$ represents complete consistency. The closer it is to 0, the better the consistency.

$$C_R = \frac{l^*}{C_I} \quad (9)$$

where the l obtained by Equation (8) is called l^* , C_I is a constant.

3.2. Objective Weight Solving Based on the Anti-Entropy Method

The entropy value is frequently mentioned when seeking objective weights in multi-criteria decision-making models. Both entropy and anti-entropy methods share a key feature: they solve for objective weights [39]. These two approaches both believe that the volatility in the data of a criterion is positively correlated with its weight. If this criterion is approximately equal for each power transmission system, then the weight corresponding to this criterion should be small. The essence of information entropy is the expected value of the amount of information.

The entropy method has been improved to become the anti-entropy method. It has the characteristic that the stronger the volatility of the criterion, the smaller the entropy value and the larger the weight coefficient [40]. The specific procedure of the anti-entropy weight method is as follows:

STEP 1: The initial matrix must be created and normalized for a comprehensive evaluation model with subjects of evaluation and criteria.

The matrix with m subjects and n criteria is $X = (x_{ij})_{m \times n}$:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix} \quad (10)$$

The initial matrix is normalized using the vector normalization method:

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}} \quad (11)$$

The matrix is normalized to eliminate the effect of the physical dimensions, and the normalized matrix R is obtained.

$$R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{bmatrix} \quad (12)$$

STEP 2: Calculate the share of the i subject under the j criterion and consider it as the probability used in the relative entropy calculation.

$$p_{ij} = \frac{x_{ij}}{\sum_{i=1}^m x_{ij}} \quad (13)$$

where p_{ij} is considered as the probability of the i subject under the j criterion.

The anti-entropy e_j is calculated in Equation (14) [25]:

$$e_j = -\frac{1}{\ln m} \sum_{i=1}^m p_{ij} \ln(1 - p_{ij}) \quad (14)$$

For each criterion, its objective weight is calculated in Equation (15) [25]:

$$\omega_j^S = \frac{1 - e_j}{n - \sum_{i=1}^n e_j}, j = 1, 2, \dots, n \quad (15)$$

where ω_j^S is the anti-entropy weight of the criterion j .

3.3. Coordination of Subjective and Objective Weights Based on Game Theory

Subjective weighting methods such as BWM depend heavily on experts' personal experiences and ideas and have some limitations. The objective weighting method is based on the data but may deviate from the actual situation. The BWM-anti-entropy weighting method is combinatorial. It is based on game theory's ideology and considers subjective and objective weights. The BWM-anti-entropy weighting method regroups the two sets of weights into a more scientific and reasonable combined weighting.

The comprehensive weighting method, which averages the two sets of weights, is unreasonable and does not effectively eliminate the shortcomings of the two weighting methods. Using game theory, this method takes Nash equilibrium as the optimum goal, coordinates the two sets of weights to find a balance, and makes the discrepancy between the final comprehensive weights and the two original sets of weights as minimal as possible to derive a set of more scientific comprehensive weights.

STEP 1: Obtain the weights of each criterion by subjective and objective methods, respectively. A basic set of weights $u = \{u_1, u_2, \dots, u_l\}$ is constructed based on this. The arbitrary linear combination of these vectors constructs the set of comprehensive weights.

$$u = \sum_{k=1}^2 g_k u_k^T, g_k > 0 \quad (16)$$

where u_1 is the subjective weight, u_2 is the objective weight, g_k and is the linear coefficient that needs to be optimized using the model.

STEP 2: Optimize g_k using a game-theoretic model that minimizes the deviation u from each u_k . The following policy response model [41] is used:

$$\min \|u - u_i\|_2 (i = 1, 2) \quad (17)$$

STEP 3: Utilize the differential property to convert the matrix constraints for the optimal solution:

$$\begin{bmatrix} u_1 u_1^T & u_2 u_1^T \\ u_2 u_1^T & u_2 u_2^T \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} = \begin{bmatrix} u_1 u_1^T \\ u_2 u_2^T \end{bmatrix} \quad (18)$$

Solve the equation $(g_1, g_2, \dots, g_l)$ normalization process:

$$g_k^* = g_k / \sum_{k=1}^2 g_k \quad (19)$$

$$u^* = \sum_{k=1}^2 g_k^* \cdot u_k^T \quad (20)$$

Optimization problems such as comprehensive weights can also be solved using Lagrange multipliers. To make the comprehensive weight closer to both objective and subjective weights, the function is established [42] as follows:

$$\min F = \sum_{j=1}^n u_j \left(\ln \frac{u_j}{u_j^*} \right) + \sum_{j=1}^n u_j \left(\ln \frac{u_j}{u_j^S} \right) \quad (21)$$

where u_j is the optimal combination weight, u_j^* is the subjective weight, and u_j^S is the objective weight.

The equation solved by the Lagrange multiplier is shown as Equation (22) [42].

$$u_j = \frac{u_j^* u_j^S}{\sum_{j=1}^n u_j^* u_j^S} \quad (22)$$

4. An Improved TOPSIS

4.1. A Solution to the TOPSIS Reverse Order Problem

The TOPSIS method constructs the multi-indicator problem's ideal and negative ideal solutions. It ranks the feasible solutions based on the evaluation criterion of the distance approximate to the absolutely ideal solution and away from the absolutely negative ideal solution.

In the traditional TOPSIS method, the reversal of order is a very prone problem for sorting multiple solutions. The root cause of the reverse order is the change in the ideal and negative ideal solutions of the decision problem after the addition of a new decision solution, which leads to a change in the evaluation criteria and, thus, to a change in the order of the superiority of the solutions [43].

If this relative ideal solution can be abandoned and absolute ideal and negative ideal solutions are defined, order reversal will not occur. More precisely, no solution is better than the absolute ideal solution or worse than the negative ideal solution in a valid decision region. Based on this concept, we define the absolutely positive ideal solution for the benefit-based criteria as follows: $S^+ = (1, 1, \dots, 1)$. Likewise, we define the absolutely negative ideal solution as follows: $S^- = (0, 0, \dots, 0)$.

4.2. Improved TOPSIS Based on Mahalanobis Distance

The Mahalanobis distance is irrelevant to the measurement scale and is not influenced by the magnitude. The covariance between criteria is added to measure the association between them. By doing so, the effect of correlation between the criteria on the distance measurement is to some extent reduced. The fundamentals of TOPSIS and the Mahalanobis distance are detailed in Appendix A.

The equation [44] for the Mahalanobis distance of A and B :

$$d(A, B) = \sqrt{(A - B)^T \Sigma^{-1} (A - B)} \quad (23)$$

where Σ is the covariance matrix.

The Mahalanobis distance only considers the covariance between the criteria. However, it solves the problem of different magnitudes between criteria and overcomes the problem of linear correlation of the criteria. However, the inverse covariance matrix only characterizes the fluctuation of the criteria compared to the mean, which may overestimate the influence between criteria, leading to inconsistent results. The Mahalanobis distance can be improved. Firstly, it would be better to use the Pearson correlation coefficient in comparison to the covariance [45]. This is because the magnitude of the criterion influences the covariance and avoids the significant differences in values that lead to poor correlation. The Pearson correlation coefficient is limited to $[-1, 1]$, which is more reasonable. Secondly, the Mahalanobis distance should be added to the comprehensive weights obtained in the previous section to calculate the distance more scientifically. In summary, the improved expression of the weighted Mahalanobis distance is as follows:

$$(A, B) = \sqrt{(A - B)^T \Delta^T \Sigma^{-1} \Delta (A - B)} \quad (24)$$

$$\Delta = \text{diag}(\sqrt{\omega_1}, \sqrt{\omega_2}, \dots, \sqrt{\omega_n})$$

where Σ is the Pearson correlation coefficient matrix, and $(\omega_1, \omega_2, \dots, \omega_n)$ are the comprehensive weights.

The steps of the improved TOPSIS comprehensive evaluation model are shown below:

STEP 1: Calculate the Pearson correlation coefficient between each criterion.

STEP 2: Conduct a normalization of different types of metrics (benefit-based, cost-based, intermediate value)

For benefit-based criteria [44]:

$$x_{ij} \begin{cases} \left(x_{ij} - \min_{1 \leq i \leq n} x_{ij} \right) / \left(\max_{1 \leq i \leq n} x_{ij} - \min_{1 \leq i \leq n} x_{ij} \right) & \max_{1 \leq i \leq n} x_{ij} \neq \min_{1 \leq i \leq n} x_{ij} \\ 1 & \max_{1 \leq i \leq n} x_{ij} = \min_{1 \leq i \leq n} x_{ij} \end{cases} \quad (25)$$

For cost-based criteria [44]:

$$x_{ij} \begin{cases} \left(\frac{\max_{1 \leq i \leq n} x_{ij} - x_{ij}}{\max_{1 \leq i \leq n} x_{ij} - \min_{1 \leq i \leq n} x_{ij}} \right) & \max_{1 \leq i \leq n} x_{ij} \neq \min_{1 \leq i \leq n} x_{ij} \\ 1 & \max_{1 \leq i \leq n} x_{ij} = \min_{1 \leq i \leq n} x_{ij} \end{cases} \quad (26)$$

For intermediate-value criteria:

$$M = \max\{|x_{ij} - x_{best}|\} \quad (27)$$

$$x_{ij} \begin{cases} 1 - \frac{|x_{ij} - x_{best}|}{M} & \max_{1 \leq i \leq n} x_{ij} \neq \min_{1 \leq i \leq n} x_{ij} \\ 1 & \max_{1 \leq i \leq n} x_{ij} = \min_{1 \leq i \leq n} x_{ij} \end{cases} \quad (28)$$

STEP 3: Calculate the distance from the i_{th} subject to the absolutely positive ideal subject $D(P_i, S^+)$ and the distance to the absolutely negative ideal subject $D(P_i, S^-)$, respectively, with Equations (29) and (30).

$$D(P_i, S^+) = \sqrt{(x_i - S^+)^T \Delta^T \Sigma^{-1} \Delta (x_i - S^+)} \quad (29)$$

$$D(P_i, S^-) = \sqrt{(x_i - S^-)^T \Delta^T \Sigma^{-1} \Delta (x_i - S^-)} \quad (30)$$

where x_i is the vector of normalized data on the criteria for the i_{th} subject, and Σ is the Pearson correlation coefficient matrix.

STEP 4: Calculate the relative closeness. A large relative closeness value responds to an economically efficient subject [44].

$$C_i = \frac{D(P_i, S^-)}{D(P_i, S^-) + D(P_i, S^+)} \quad (31)$$

5. Example Analysis and Comparison

Transmission and transformation projects are the basic projects for the realization of the transmission system, which is responsible for the construction and maintenance of the lines and facilities required for the transmission system. This section analyzes data related to ten 500 kV transmission and transformation projects. These projects are comprehensively evaluated using the improved TOPSIS and BWM-anti-entropy weight method. First, the subjective and objective weights are sought, and then the comprehensive weights are obtained by applying game theory. Finally, a comprehensive evaluation of the ten projects is carried out based on the calculated comprehensive weights and the improved TOPSIS method.

5.1. Weighting Solution

5.1.1. Subjective Weights

The preferences of each criterion concerning the best and worst criterion are calculated in the order of 1 to 9, according to the order of comparing the best criterion with the other indicators and the other indicators with the worst criterion.

Select the Best: Return on Investment

Select the Worst: Profit Growth Rate

The specific scores are detailed in Table 4.

Table 4. Scoring the relative importance of the best and worst criteria.

	C_1	C_2	C_3	C_4	C_5	C_6	C_7
C_B	1	2	3	2	5	6	7
C_W	7	6	4	5	3	3	1

The results of the consistency test calculated according to the formula are:

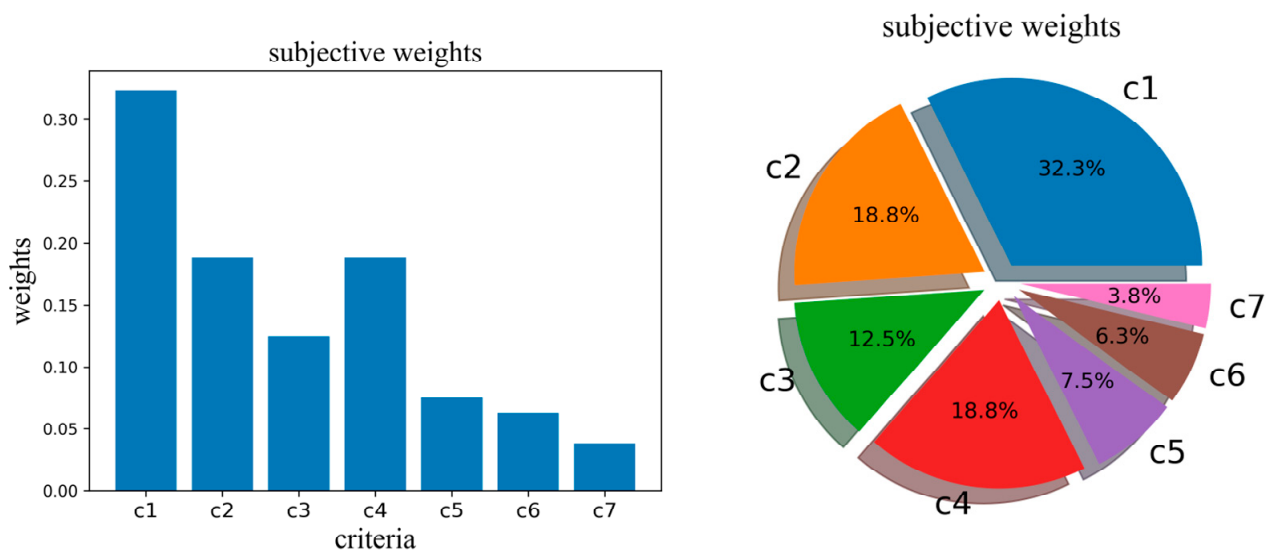
$$\begin{aligned} \text{Input-Based, } CR &= 0.261 \\ \text{Associated Threshold} &= 0.314 \end{aligned}$$

$CR < \text{Associated Threshold}$; so, the pairwise comparison consistency level is acceptable.

After solving the mathematical planning, the optimal weights of BWM are shown in Table 5. The percentages and comparisons for the criteria are shown in Figure 3.

Table 5. Subjective weights.

Subjective Weights	C_1	C_2	C_3	C_4	C_5	C_6	C_7
	0.323	0.188	0.125	0.188	0.075	0.063	0.038

**Figure 3.** Subjective weights.

5.1.2. Objective Weights

In this paper, anti-entropy weighting is applied for the objective weighting solution. This method derives its principle from the coefficient of variation and entropy weights.

Seven criteria are normalized for ten 500 kV transmission projects. The normalized matrix is detailed in Table 6.

The anti-entropy and weights of each criterion obtained according to the relevant formula of the anti-entropy weighting method are listed in Table 7.

The weights and anti-entropy derived from the anti-entropy weights method can be seen: Net Present Value and Return On Investment have higher volatility and thus higher anti-entropy and weights. The data of other criteria do not change too much, and the weights are within a reasonable range.

Table 6. Vector normalization.

Project Code	Return on Investment	Net Present Value	Payback Period	Internal Rate of Return	Asset Liability Ratio	Current Ratio	Profit Growth Rate
P_1	0.3532	0.1591	0.3737	0.2903	0.3157	0.3074	0.4416
P_2	0.3532	0.1120	0.1925	0.3736	0.2894	0.2886	0.3865
P_3	0.5564	0.0343	0.2864	0.4885	0.2736	0.4811	0.2562
P_4	0.3263	0.2968	0.3251	0.3699	0.3262	0.3098	0.2542
P_5	0.2523	0.1248	0.2459	0.2698	0.3315	0.2746	0.2767
P_6	0.3499	0.0724	0.3121	0.3535	0.3262	0.3027	0.4336
P_7	0.2187	0.2103	0.4262	0.1351	0.2421	0.3520	0.2629
P_8	0.1817	0.2202	0.3728	0.1609	0.2999	0.2534	0.1256
P_9	0.2052	0.3472	0.3013	0.2799	0.3052	0.2746	0.2334
P_{10}	0.1591	0.7232	0.2575	0.2836	0.4210	0.2534	0.3485

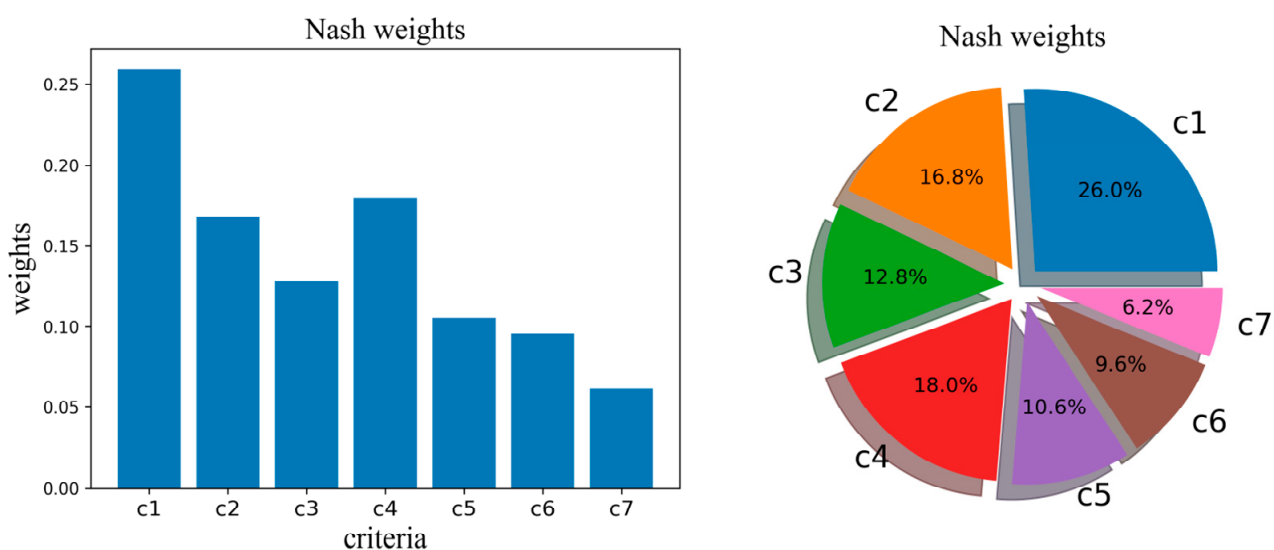
Table 7. Anti-entropy and subjective weights.

	Return on Investment	Net Present Value	Payback Period	Internal Rate of Return	Asset Liability Ratio	Current Ratio	Profit Growth Rate
Anti-entropy	0.0533	0.0771	0.0480	0.0512	0.0468	0.0480	0.0507
Weights	0.1475	0.1510	0.1391	0.1446	0.1379	0.1391	0.1396

5.1.3. Optimal Comprehensive Weights

Neither subjective nor objective weights can fully cover the information provided by the criteria. This section calculates the comprehensive weights using the Nash equilibrium and Lagrange multiplier methods.

According to the game model given by the Nash equilibrium in this paper, the coefficients of the combination of subjective and objective weights, which satisfy the Nash equilibrium, are 0.2241 $g_2 = 0.7758$. The combined Nash weights obtained by combining the two weights by the coefficients: $u^* = \{0.26, 0.17, 0.12, 0.18, 0.11, 0.10, 0.06\}$. The percentages and comparisons for the criteria are detailed in Figure 4.

**Figure 4.** Nash weights.

The Lagrange multiplier method can also solve the problem of finding the extrema under a set of constraints. Solving the objective function with the Lagrange multiplier method

yields the optimal combination of weights: $u^* = \{0.27, 0.17, 0.13, 0.18, 0.11, 0.09, 0.05\}$. A comparison of the two sets of basic weights (subjective, objective) and the two sets of combined weights (Nash, Lagrange) is shown in Figure 5.

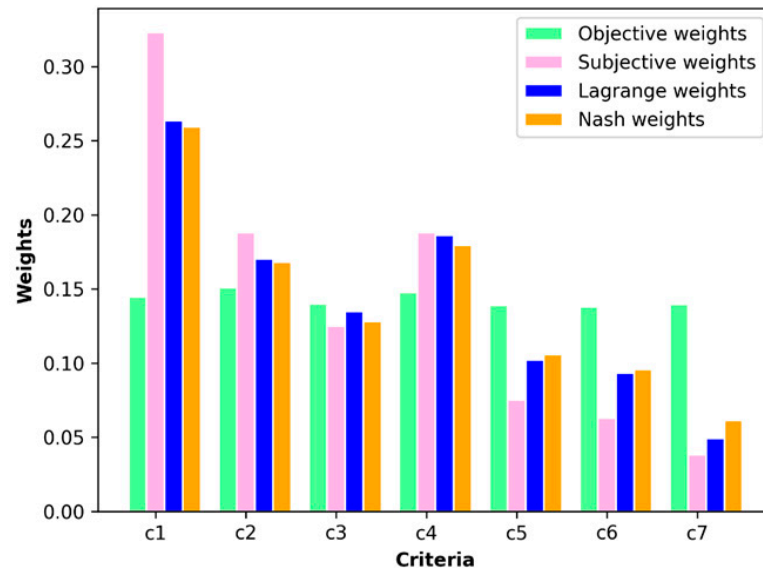


Figure 5. Comparison of different weighting methods.

This section solves the comprehensive weights in two separate approaches. Comparing the two combined weights, the Nash and Lagrange weights are very similar, and both are between the subjective weights and objective weights. In this paper, Nash weight is chosen as the comprehensive weight of BWM-anti-entropy.

5.2. An Improved TOPSIS Method to Calculate Score Evaluation

The Pearson correlation coefficient matrix between the seven criteria Σ is shown in Figure 6.

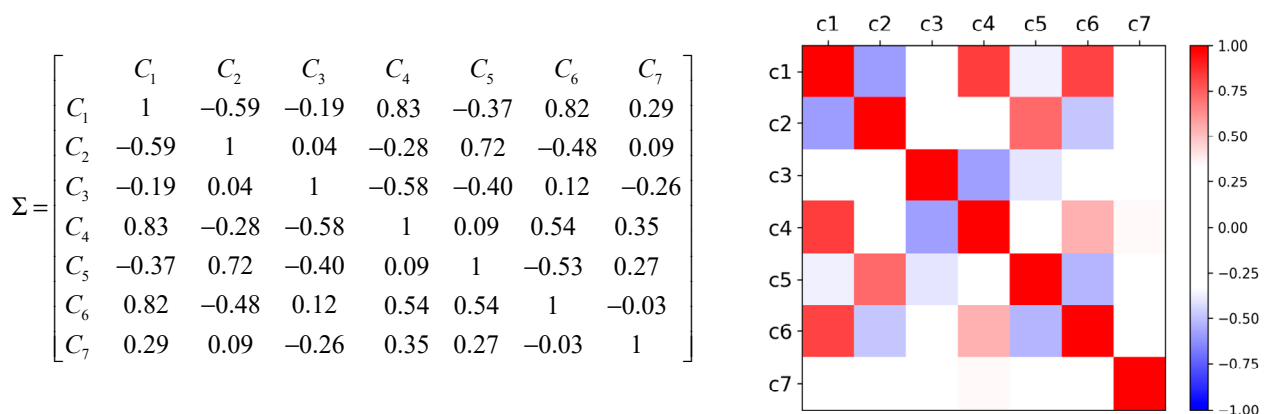


Figure 6. Pearson correlation coefficient matrix.

As can be seen from this Pearson correlation coefficient matrix, some of these criteria are strongly correlated. If we continue to use the Euclidean distance, it will lead to the failure of the Euclidean distance and will be unable to correctly assess the distance from the actual projects to the ideal projects. At the same time, using the inverse Pearson coefficient matrix as a substitute for the inverse covariance matrix will reduce the influence of the magnitude to a certain extent and more reasonably measure the degree of correlation between the criteria.

The data-processed results in advance by Max-Min normalization are detailed in Table 8.

Table 8. Max-Min normalization.

Project Code	Return on Investment	Net Present Value	Payback Period	Internal Rate of Return	Asset Liability Ratio	Current Ratio	Profit Growth Rate
P_1	0.4886	0.4953	0.2245	0.4390	0.5882	0.2500	1.0000
P_2	0.4886	0.1128	1.0000	0.6748	0.7353	0.1630	0.8258
P_3	1.0000	0.0000	0.5984	1.0000	0.8235	1.0000	0.4136
P_4	0.4208	0.3811	0.4325	0.6642	0.5294	0.2609	0.4071
P_5	0.2345	0.1313	0.7714	0.3813	0.5000	0.0978	0.4784
P_6	0.4801	0.0553	0.4882	0.6179	0.5294	0.2283	0.9747
P_7	0.1499	0.2555	0.0000	0.0000	1.0000	0.4565	0.4346
P_8	0.0567	0.2699	0.2286	0.0732	0.6765	0.0000	0.0000
P_9	0.1160	0.4541	0.5345	0.4098	0.6471	0.0978	0.3412
P_{10}	0.0000	1.0000	0.7219	0.4203	0.0000	0.0000	0.7055

The comprehensive weights and Pearson correlation coefficient matrices obtained from the Nash game model are used in TOPSIS to solve for the Mahalanobis distance.

The equation is used to find the Mahalanobis distance and the relative closeness of the evaluated solutions to the absolutely positive ideal projects and the negative ideal projects, and the evaluation projects are ranked according to the closeness. High relative closeness indicates that the project should be good. The Mahalanobis distance and closeness ranking are shown in Table 9.

Table 9. Mahalanobis distance and closeness ranking.

	Distance of Positive Solution	Distance of Negative Solution	Relative Closeness	Rank
P_3	0.6934	1.1312	0.6200	1
P_1	0.7582	0.7644	0.5020	2
P_2	1.1716	1.0347	0.4690	3
P_{10}	1.6661	1.1719	0.4129	4
P_4	0.9828	0.6726	0.4063	5
P_6	1.0899	0.6780	0.3835	6
P_7	1.3983	0.8604	0.3809	7
P_5	1.2656	0.7283	0.3653	8
P_9	1.3010	0.6603	0.3367	9
P_8	1.1567	0.4296	0.2708	10

After comparing the data with the final distance and closeness in Table 9, it can be seen that the relative closeness of P_3 is the highest. This is mainly because the two criteria with the highest weights, Return on Investment and Internal Rate of Return, are optimal. In contrast, for the cost-based criterion Asset Liability Ratio, P_1 is low. For P_8 , all four criteria characterizing profitability (c1~c4) are low, leading to poor evaluation results.

5.3. SIMUS and Traditional TOPSIS Solving

So far some of the more popular MCDM methods are AHP, TOPSIS, SPOTIS, COMET, SIMUS, etc. The SPOTIS method no longer performs relative comparisons between projects to avoid ranking reversal and forms a stable score preference ranking [43]. The core idea of COMET (characteristic object method) is to assign a score or ranking to each object by comparing the similarity between the object and a set of characteristic objects such as the Euclidean distance and cosine similarity [46]. COMET needs more information. The SIMUS method is not based on the same principles as these methods. It uses a linear programming algorithm to generate the Pareto efficient matrix, i.e., efficient results matrix,

thus obtaining an optimal solution (scores) for each criterion. The Pareto efficient matrix is analyzed vertically, and the total score and ranking of each project is found based on the weights of the objective data (efficient results matrix ranking). The method then re-scored each project from a horizontal analysis perspective based on the ERM, forming a new matrix called the project dominant matrix (PDM). Even though the scores generated from the two perspectives are different, the ranking of the corresponding scores from the two perspectives is the same and stable.

In this paper, the traditional TOPSIS method and SIMUS method are used for comprehensive economic evaluation, respectively. The results are eventually compared with those of the proposed improved TOPSIS based on BWM-anti-entropy weight to demonstrate the feasibility of the method.

5.3.1. Traditional TOPSIS Solving

The traditional TOPSIS method considers the weights and determines the optimal and worst projects based on the decision matrix, immediately followed by the calculation of the Euclidean distance and closeness ranking.

The normalized matrix is detailed in Table 6. The optimal comprehensive weights are determined as Nash weights calculated by Section 5.1.3. The Euclidean distance and closeness ranking are shown in Table 10.

Table 10. Euclidean distance and closeness ranking.

	Distance of Positive Solution	Distance of Negative Solution	Relative Closeness	Rank
P_{10}	0.2403	0.3004	0.5556	1
P_3	0.2876	0.2722	0.4862	2
P_4	0.2334	0.1796	0.4349	3
P_2	0.2821	0.1846	0.3956	4
P_9	0.2681	0.1578	0.3704	5
P_1	0.2805	0.1596	0.3588	6
P_6	0.3015	0.1640	0.3523	7
P_5	0.3152	0.1155	0.2681	8
P_7	0.3262	0.1076	0.2481	9
P_8	0.3366	0.0892	0.2096	10

5.3.2. SIMUS Solving

The sequential interactive model for urban systems (SIMUS) is a model developed to support urban planning and management decisions and is now mostly used in MCDM. Based on linear programming, SIMUS is immune to rank reversal [47]. An important feature of the SIMUS model is its sequential interactivity, which means that the model runs as an iterative process, with each step based on the results of the previous step.

In this study, SIMUS V.3.13, an open-source software that has a vast user community [48], is used for evaluation to determine the optimal or most viable solution. The specific steps are as follows:

STEP 1: Create a project and enter the number of criteria and alternatives as shown in Figure 7.

STEP 2: Create and normalize a decision matrix. The results were the same as in Table 6.

STEP 3: Start analysis, then export the efficient result matrix (ERM) and the project dominance matrix (PDM). The results are detailed in Tables 11 and 12.

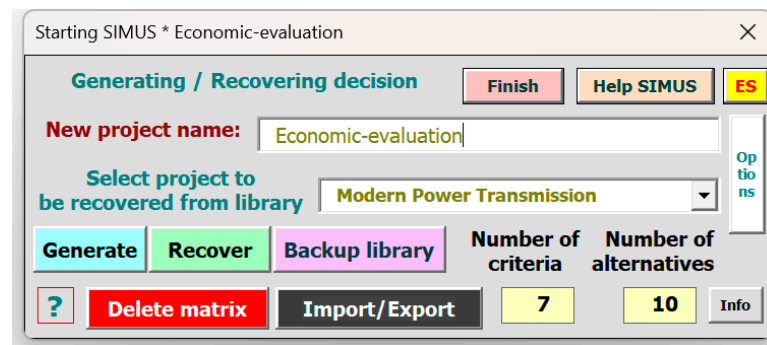


Figure 7. Criteria and alternatives in SIMUS.

Table 11. Normalized efficient result matrix.

	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9	P_{10}
C_1		1.00								
C_2										1.00
C_3				0.27						0.73
C_4		0.81							0.19	
C_5							1.00			
C_6			0.29				0.63	0.08		
C_7	0.65					0.35				
Sum of Column (SC)	0.65	1.81	0.29	0.27	0.00	0.35	1.63	0.08	0.19	1.73
Participation Factor (PF)	1	2	1	1	0	1	2	1	1	2
Norm. Participation Factor (NPF)	0.14	0.29	0.14	0.14	0.00	0.14	0.29	0.14	0.14	0.29
Final Result (SC $\times$ NPF)	0.09	0.52	0.04	0.04	0.00	0.05	0.46	0.01	0.03	0.49

Table 12. Project dominance matrix.

	P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9	P_{10}	Row Sum	Net
P_1		0.65	0.65	0.65	0.65	0.30	0.65	0.65	0.65	0.65	5.50	−0.50
P_2	1.81		1.81	1.81	1.81	1.81	1.81	1.81	1.62	1.81	16.1	11.1
P_3	0.29	0.29		0.29	0.29	0.29		0.21	0.29	0.29	2.24	−4.01
P_4	0.27	0.27	0.27		0.27	0.27	0.27	0.27	0.27		2.16	−4.30
P_5											0.00	−7.00
P_6		0.35	0.35	0.35	0.35		0.35	0.35	0.35	0.35	2.80	−3.50
P_7	1.63	1.63	1.34	1.63	1.63	1.63		1.55	1.63	1.63	14.3	9.30
P_8	0.08	0.08		0.08	0.08	0.08			0.08	0.08	0.56	−6.20
P_9	0.19		0.19	0.19	0.19	0.19	0.19	0.19		0.19	1.52	−5.10
P_{10}	1.73	1.73	1.73	1.46	1.73	1.73	1.73	1.73	1.73		15.3	10.3
Column sum	6.00	5.00	6.34	6.46	7.00	6.30	5.00	6.76	6.62	5.00		

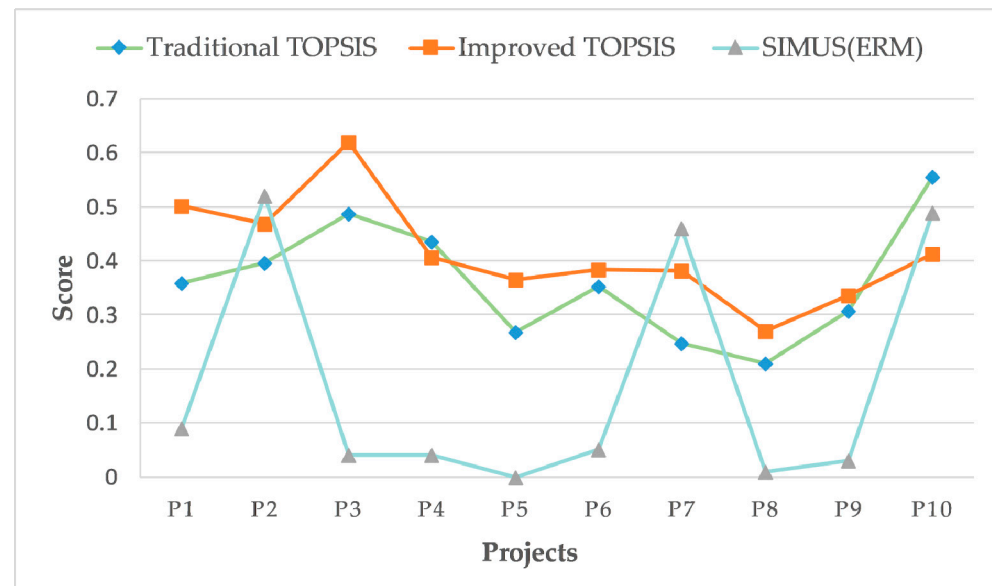
Final Result in ERM and NET in PDM are the scores of 10 projects from two perspectives based on linear programming and mathematical calculations by the SIMUS method. A high score represents a suitable program.

5.3.3. Comparative Analysis of Evaluation Result

All four sets of scores (traditional and improved relative closeness, NET, and Final Result) were obtained by three different methods or perspectives to rank the example projects, as shown in Table 13. The rankings obtained by the ERM and the DPM are identical. A comparison of the results of the three methods is shown in Figure 8.

Table 13. Comparison of evaluation scores and rankings.

Rank	TOPSIS Method				SIMUS Method			
	Traditional TOPSIS Project	Traditional TOPSIS Score	Improved TOPSIS Project	Improved TOPSIS Score	ERM Project	ERM Score	PDM Project	PDM Score
1	P_{10}	0.5556	P_3	0.6200	P_2	0.52	P_2	11.1
2	P_3	0.4862	P_1	0.5020	P_{10}	0.49	P_{10}	10.3
3	P_4	0.4349	P_2	0.4690	P_7	0.46	P_7	9.3
4	P_2	0.3956	P_{10}	0.4129	P_1	0.09	P_1	−0.5
5	P_9	0.3704	P_4	0.4063	P_6	0.05	P_6	−3.5
6	P_1	0.3588	P_6	0.3835	P_3	0.04	P_3	−4.01
7	P_6	0.3523	P_7	0.3809	P_4	0.04	P_4	−4.3
8	P_5	0.2681	P_5	0.3653	P_9	0.03	P_9	−5.1
9	P_7	0.2481	P_9	0.3367	P_8	0.01	P_8	−6.2
10	P_8	0.2096	P_8	0.2708	P_5	0	P_5	−7

**Figure 8.** Comparison of evaluation scores.

Differing methods often lead to varying rankings and preferences in MCDA. Using appropriate coefficients to test the similarity of the rankings would promote a better, more responsible decision [49]. ρ Spearman, τ Kendall, and γ Goodman-Kruskal coefficients are commonly used to check the similarity of rankings [50].

In the comparison analysis using traditional coefficients, the effect of the placement of a pair of adjacent alternatives on similarity is the same for each position in the ranking. In rank correlation analysis using traditional coefficients, interchanging the top two rankings has a commensurate effect to swapping the bottom two on the outcome [51]. This is not rational in MCDA. Hence, a new coefficient WS is proposed to compare the similarity between the rankings, making the differences within the top-ranked intervals more significant [52]:

$$WS = 1 - \sum_{i=1}^n \left(2^{-R_{xi}} \cdot \frac{|R_{xi} - R_{yi}|}{\max\{|1 - R_{xi}|, |N - R_{xi}|\}} \right) \quad (32)$$

where WS is a value of the similarity coefficient, N is a length of ranking, and R_{xi} and R_{yi} mean the place in the ranking for i -th element in respectively ranking x and ranking y . The result of the WS coefficient is detailed in Table 14.

Table 14. WS coefficient between the three rankings.

WS	Traditional TOPSIS	Improved TOPSIS	SIMUS Method
Traditional TOPSIS	1.0000	0.7167	0.6865
Improved TOPSIS	0.7167	1.0000	0.5809
SIMUS Method	0.6865	0.5809	1.0000

The comparison results are shown in Tables 13 and 14. The rankings of all three methods have some relatively strong correlation, which follows common sense. The higher correlation between traditional TOPSIS and improved TOPSIS may be because they both use the composite weights derived in the previous section, whereas the SIMUS method focuses on iterating to find the best in the objective data through linear programming. The highest score in the SIMUS method is also because P_2 performs well in C_5 and C_7 . The weights of C_5 and C_7 in comprehensive weights are very small, so P_2 is not ranked as high in the TOPSIS method.

It can be seen in Table 13 that there are some deviations in the ranking results formed by the improved TOPSIS and the traditional TOPSIS. For example, the rankings of P_1 and P_9 in the two sets of results differ considerably, which is due to the more significant correlation in the indicator system. The improved TOPSIS method is not affected by correlation and magnitude. In contrast, the traditional Euclidean distance is affected by correlation, which leads to the problem of reversed ordering.

In summary, both the individual and comparative analyses have effectively demonstrated the rationality and feasibility of the improved TOPSIS method based on the weighted Mahalanobis distance and the absolute ideal solution proposed in this paper in the economic evaluation of power systems.

6. Conclusions

This paper proposes a comprehensive evaluation method based on improved TOPSIS and BWM-anti-entropy weighting. It applies to the power transmission system, which benefits the investment decision and management. The conclusions are as follows:

1. The economic comprehensive evaluation model enhances the tedious process of the traditional AHP method of two-by-two comparison of criteria, and the consistency and reliability of the results are higher. While adding objective weights, the optimal coordination of weights is solved by the Nash equilibrium and Lagrange multiplier methods, respectively. Weights become more accurate and reasonable.
2. Improving the TOPSIS method using the Mahalanobis distance and comprehensive weights solves the problem of Euclidean distance failure and inverse order when the linear correlation between criteria is high. Applying the Pearson correlation coefficient rather than covariance circumvents the problems caused by covariance's inconsistency in magnitude between criteria.
3. A comprehensive evaluation of the power transmission system economy was achieved by ranking the relative closeness of ten 500 kV transmission projects to the positive and negative ideal projects. The economic evaluation was performed using the traditional TOPSIS, improved TOPSIS, and SIMUS methods for economic integration, respectively. A comparative analysis was implemented to analyze the differences between the three rankings and their causes. The results indicate that this method provides both an economic evaluation of the power transmission systems and a reflection of the closeness of each criterion to the ideal situation. The evaluation criteria system is characterized by its comprehensiveness and objectivity, while the evaluation process is conducted scientifically and reasonably. As a result, the final evaluation conclusion possesses a high degree of credibility.

Based on comprehensive weights, the method improves the TOPSIS by introducing the Mahalanobis distance and Pearson correlation coefficients, which can eliminate the influence of linear correlation. This method is essentially a multi-attribute decision-making

(MADM) method that breaks through the limitations of single-dimension weights. The evaluation can also be expanded to include safety factors and sensitivity, etc. The research object can also be expanded to an entire power grid in a region. Hence, this method is widely applied and can be implemented in the economic analysis of additional diverse subjects. The TOPSIS method will be extended to multi-attribute group decision-making issues.

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Appendix A

As shown in Figure A1, using two dimensions as an example, point B is apparently more of an outlier than point A from a statistical point of view. Nevertheless, if we calculate the Euclidean distance of these two points to the center O of the point group, point A is more out of the group. This unreasonable conclusion occurs because there is a linear correlation between x_1 and x_2 . To avoid the negative influence of this correlation, the coordinate system can be rotated counterclockwise by a certain angle into a new coordinate system y_1Oy_2 .

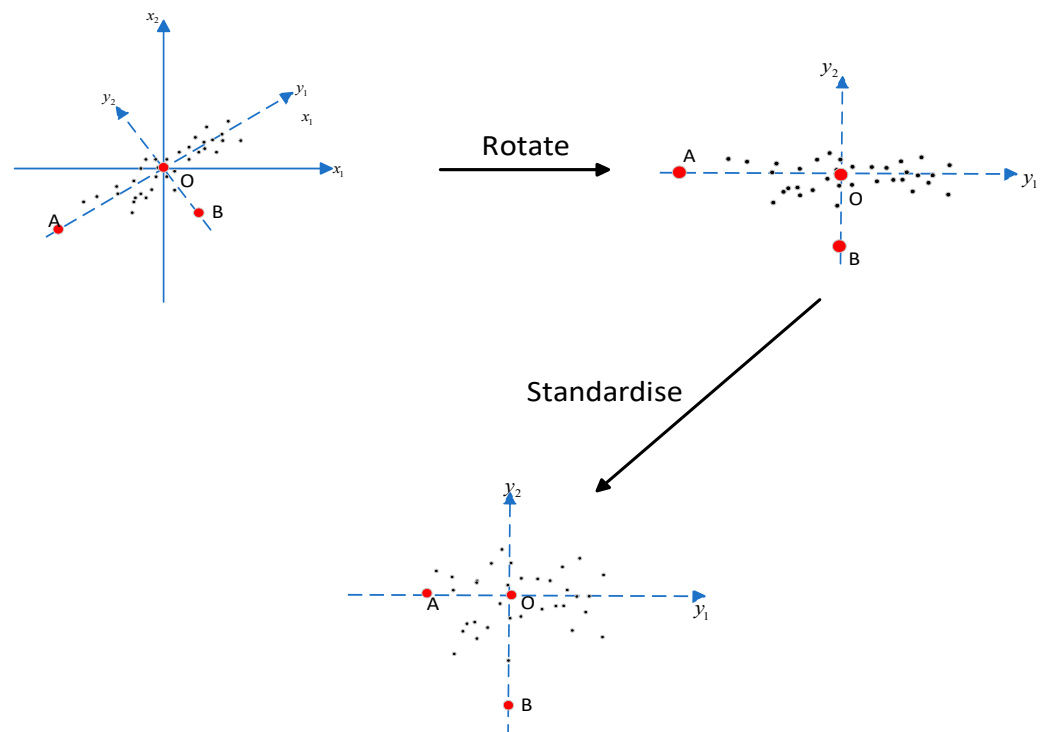


Figure A1. Mahalanobis distance.

Under the new coordinate system y_1 and y_2 are uncorrelated, and, after the normalization transformation, the elliptical point group becomes a circular point group. Currently, the Euclidean distance calculated according to the formula is the Mahalanobis distance. It

is not difficult to see that the Mahalanobis distance from point A to point O is shorter than the Mahalanobis distance from point B to point O .

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